Charu Bhat

FPGA Design Engineer

RESUME_1910

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Total Experience: Fresher; internship experience

PROFESSIONAL SUMMARY

Entry-level FPGA Design Engineer trained in VHDL, Verilog, Xilinx Vivado, Intel Quartus. Has hands-on academic and internship experience in rtl simulation, implementation, test documentation and engineering problem solving.

TECHNICAL SKILLS

Skills: FPGA, Python, VHDL, Intel Quartus, Tcl, AXI4, Xilinx Vivado, Verilog, DSP, Timing Closure, Hardware Simulation, ILA

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Engineering Intern - FPGA Design | HelioNex Labs | Jan 2026 - Jun 2026

- Lowered end-to-end data-path latency by 31% using pipelined logic
- Wrote repeatable test notes, reviewed results with a mentor and documented prioritized fixes.
- Demonstrated prototype trade-offs using fixed-point arithmetic.

PROJECTS

Protocol Adapter Core | Tcl, Xilinx Vivado, VHDL, Python

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Real-Time Signal Filter | AXI4, Tcl, Xilinx Vivado, VHDL

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Blue River College of Technology, 2026
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Timing Closure Training (2026)